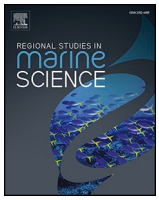
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Cigarette butts on neotropical beaches with different levels of development

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ABSTRACT

The study aims to understand the spatial and temporal distribution of cigarette butts in different areas of beaches, as well as to compare two collection methods and propose this waste as an indicator item to assess beach contamination. Cigarette butts were collected in different locations (space), in the Solarium Area and the High Tide Line Area on beaches with different levels of development, such as Boa Viagem (high), Porto de Galinhas (medium), and Cupê (low), and also at different times (temporality), beginning of rain, rain, beginning of summer, and summer. In the Solarium Area, two methods were applied in five 1.5 m wide transects at 10 m intervals, while in the High Tide Line Area, only one method was used in three 100 m transects at 100 m intervals. The data were analyzed according to the cigarette butts found, their level of degradation, morphometry, brand audit, and contamination index, which was estimated based on the density of the butts and environmental factors. Adding up the areas of the Solarium and High Tide Line, 2636 cigarette butts were collected, of which 52 % were found on the highly developed beach (BV), 40 % on the moderately developed beach (PG), and 8 % on the poorly developed beach (CP). The equipment showed a greater capacity to capture cigarette butts, with a density of 1.87 butts/m^2, compared to manual collection, which obtained a value of 1.33 butts/m^2. In addition, the degradation of cigarette butts influenced environmental pollution, with a higher density of butts in an advanced stage of degeneration.

1. Introduction

Cigarette butts are composed of cellulose acetate, a synthetic polymer derived from cellulose with great commercial importance (Araújo and Costa, 2019). This material is highly durable and decomposes slowly in the environment (Puls et al., 2011). Due to their high filtration capacity, the compacted polymer fibres present in cigarette butts end up accumulating toxic substances that can be leached into the environment. This contamination process affects various ecosystems and can result in the transport and accumulation of nanoparticles in soils and the ocean (El Hadri et al., 2021).

Urban beaches with high recreational use are heavily contaminated with plastic items, including cigarette butts (Silva, Araújo and Silva-Cavalcanti, 2023a; Araújo and Costa, 2019; Silva et al., 2008). Estimates made by the Ocean Conservancy (2025), through international coastal clean-up events, showed that cigarette butts ranked first in the number of items most commonly found. It is estimated that 5.5 trillion cigarettes are consumed globally each year, with around 2/3 of cigarette butts dumped in public places by smokers (Novotny and Slaughter, 2014), including beaches. The high correlation between irregular disposal and urbanization, as well as tourist activities, reinforces the need to adopt different methods of capturing and removing this waste in order to mitigate its environmental impacts. SDG-14 proposes the conservation of the oceans, preventing and reducing marine pollution, with a target of 2030 (United Nations, 2025).

Areas with major economic and environmental interests, such as Boa Viagem beach and Porto de Galinhas, both considered Environmental Protection Areas (ISA – Instituto Socioambiental, 2025), require greater care and monitoring of the irregular disposal of cigarette butts. Because, even without tobacco remains, cigarette butts contain chemical substances adsorbed on their fibres, which can be spread in the environment, becoming a source of toxicity causing irreversible damage to various organisms (Bonanomi et al., 2020; Slaughter et al., 2011; Lima et al., 2021; Montalvão et al., 2019).

Understanding the spatial and temporal distribution of cigarette butts is fundamental to identifying their main sources and sinks. As this waste can behave similarly to microplastics, accumulating, fragmenting and adsorbing organic pollutants, its detailed monitoring is necessary

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for the development of effective management strategies. In this sense, surveying the density of cigarette butts both on the sand strip and in the High Tide Line Area makes it possible to understand patterns of dispersion and accumulation, guiding the adoption of more appropriate measures to control them.

Given this scenario, we propose using cigarette butts as a flag item to assess contamination levels on beaches in different contexts. The objective of this study is to conduct a survey of this solid waste in coastal environments with varying degrees of development, understand its spatial distribution in different regions of the beach (Solarium Area and High Tide Line Area), and compare the efficiency of two different sampling methods for capturing this contaminant.

2. Material and methods

2.1. Study areas

To make it possible to monitor cigarette butts on beaches with different characteristics, a preliminary study was carried out in May 2023 using the parameters described in Supplementary Table 1. Defining study areas with different patterns was essential to understanding the types and magnitude of the impacts caused by human action on environmental quality, considering the inappropriate disposal of cigarette butts (Table 1).

As a result, it was established that the beaches of Cupe (8º27'18.57''S, 34º59'03.56''O) and Porto de Galinhas (8º30'12.12''S, 35º00'05.20''O), both located in the municipality of Ipojuca; and Boa Viagem which is situated in the municipality of Recife (8º07'04.95''S, 34º53'36.24''O) have their levels of development and use established as low, medium and high, respectively (Fig. 1).

According to IBGE data (2025), the municipality of Ipojuca, located in the metropolitan mesoregion of Recife, is home to Porto de Galinhas beach and Cupe beach, both noted for their tourist activities. However, when comparing the territorial extension of these beaches, there is a difference between the demographic data (Table 2). Porto de Galinhas has a coastline of 18 km, while Cupe beach stretches for just 4 km. In Recife, Boa Viagem beach accounts for 72 % of the municipality's coastline, totalling 11 km of shoreline.

2.2. Collection of cigarette butts

For both regions, the Solarium and the High Tide Line Area, collections were made at low tide during the spring tides (full moon), with collections made at the start of the rainy season (March), the rainy season (June), the start of the dry season (September) and the dry season (December) of 2024. In addition, to ensure better standardisation, the field trips always took place on the same day of the week, with collections in Cupe and Porto de Galinhas taking place on Mondays and in Boa Viagem on Tuesdays.

Table 1
Three classes are defined to categorise stretches of waterfront, according to Projeto (2002).

Classification of use and development		
High	Medium	Low
Ecosystems already impacted, with medium or high population density. The landscape elements, the physical support and the type of urbanization are essential factors in classifying the waterfront.	It contains partially altered ecosystems, with areas of low to medium population density.	It is characterised by preserved ecosystems with low occupation.

2.2.1. Solarium area

The approach used to collect cigarette butts was adapted by Silva et al., 2023b establishing transects positioned in the Solarium Area (Polette and Raucci, 2003). This area of the Solarium is a region within the sandy strip where there are people and movement of users who remain on the beach. In order to compare the success of the cigarette butt capture, two sampling methods were applied to the sand strip: (1) the first consisted of a manual search, in which the material present on the surface of the sand strip was removed, up to 2 cm deep from the surface; (2) the second method used a beach cleaning equipment model *Sieve Solid Waste Collector Beach Tennis Sand.* For each sampling method, five 1.5 m wide transects were established at 10 m intervals, positioned up to the high tide line. In addition, the transects were grouped in pairs so that collections with the different methods took place in the same area of the sand strip (Fig. 2).

2.2.2. High tide line area

In order to compare the areas occupied by beach users, cigarette butts were also collected in the High Tide Line Area using the method adapted from Silva et al. (2008), where 3 transects 100 m long and 1.5 m wide were positioned at 100 m intervals from High Tide Line Area). Samples were taken manually, as this region is highly compacted, making it impossible to use beach cleaning equipment.

2.3. Sorting cigarette butts

After collection, the material was dried in an oven at 60º C until it reached a constant weight, to avoid any interference caused by water retention. They were then weighed and morphometrically measured, including length and width, to determine their mass. A brand audit was also carried out, with the aim of identifying the predominant polluters on the beaches. In addition, the 4 levels of degradation of cigarette butts were observed: Type I: cigarette butt with preserved filter and paper coating, containing tobacco; Type II: preserved filter and coating, but no tobacco present; Type III: worn paper coating (still present), no tobacco; Type IV: only cellulose acetate, no tobacco and no paper coating, as determined by Araújo et al. (2022) and shown in Fig. 3.

2.4. Beach contamination index

The studies carried out by Torkashvand et al. (2021) make it possible to determine the contamination index of urban areas by calculating "CBPI = Cigarette Butt Pollution Index x E", with 'E' being the combination of several variables, (1) such as soil, (2) rainfall, (3) urban roads and (4) proximity to groundwater. However, no data was found on the types of (3) urban trails and the (4) relationship with groundwater for the study area, so we carried out the calculation without these variables, but we also made a comparison with another study that assessed contamination by visual perception through the number of items/m^2, using plastic debris as an indicator of beach cleanliness (Alkalay, Pasternak and Zask, 2007), and we performed the comparison in the current study with cigarette butts. The reference values (Table 3) can be used to establish the level of contamination through the application of different studies to potentialize the harmful effects of cigarette butts on beaches.

2.5. Analysing data

ANOVA was applied to analyse significant differences ($p < 0.05$) between areas with different occupancy configurations, collections, methods, transects, levels of degradation and morphometry when the data showed a normal distribution. Otherwise, the Kruskal-Wallis test was used, both carried out in the BioEstat 5.4 software. In addition, to assess the correlation between two variables, Pearson's linear correlation coefficient was used to observe the relationship between the rainfall index and the distribution of cigarette butts at different times of the year. The amount of cigarette butts was standardized as the number of

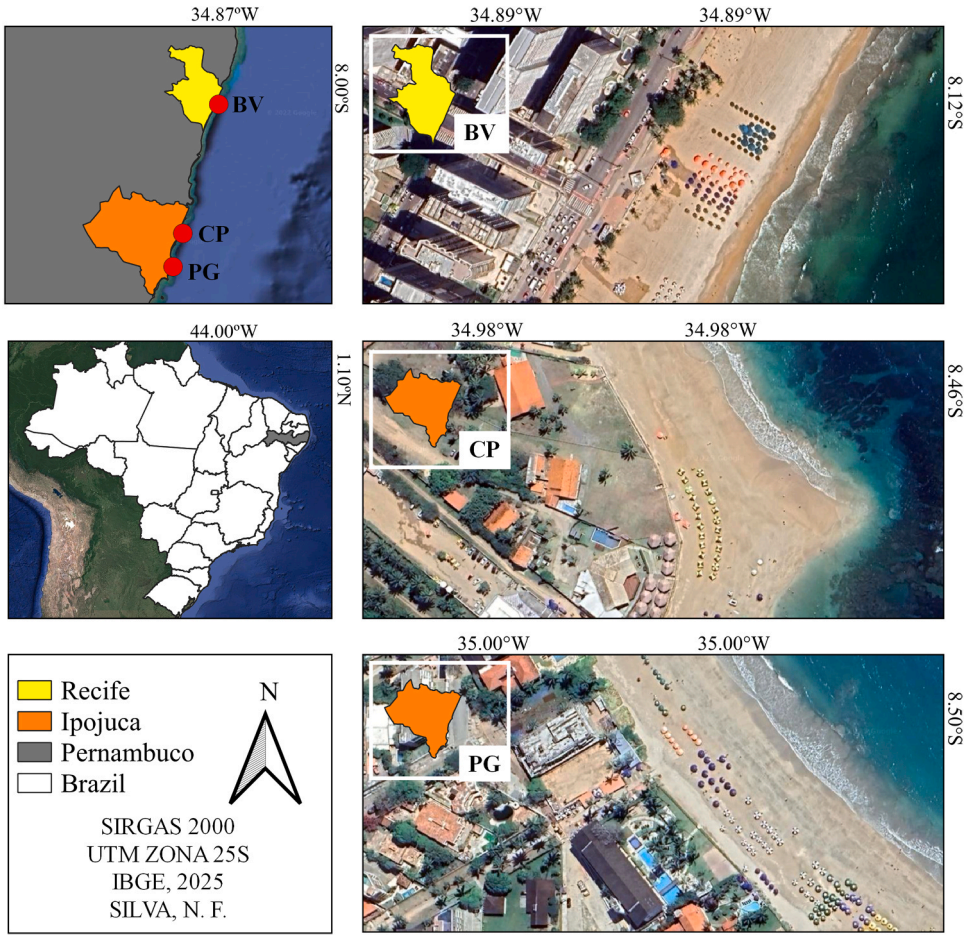


Fig. 1. Map of the points on the Pernambuco coast, the cigarette butt collections were carried out on Boa Viagem beach, with a high level of development ● (BV), Porto de Galinhas beach, with a medium level of development ● (PG) and Cupe beach, with a low level of development ● (CP). Source: Nirhvana Felipe, 2025.

Table 2
Information on the municipalities where the beaches used as catchment areas for beaches with high levels of development (BV), medium levels of development (PG), and low levels of development (CP) are located, with information based on IBGE - Instituto Brasileiro De Geografia E Estatística (2025).

Municipality	Population/ km²	Urbanized area	Coastal extension	Beach extension
Recife	6803.60	142.99 km²	11 km	BV = 8 km
Ipojuca	189.60	29.39 km²	32 km	PG = 18 km CP = 4 km

cigarette butts/m², given that the regions (Solarium Area and High Tide Line Area), as well as the transects, have different dimensions.

3. Results and discussion

In total, adding the Solarium Areas and the High Tide Line Area, 2636 cigarette butts were captured, of which 52 % were found on the beach with high development (BV), 40 % on the beach with medium development (PG) and 8 % on the beach with low development (CP) (Fig. 4), with a significant difference ($p < 0.05$). The densities in these areas varied between the beaches, however, the beaches with high and medium development did not differ significantly ($p > 0.05$), with values close to each other, where: BV obtained 4.91 cigarette butts/m² and PG with 3.84 cigarette butts/m². In CP, there was a drop, with a significant difference ($p < 0.05$) with values of 0.71 cigarette butts/m².

According to Chen (2018), the rate of waste is higher than the rate of urbanization, and the problem may be even greater than that recorded. In addition, the density of cigarette butts, similar to that observed on beaches with high and medium development, may be related to both the number of visitors and the presence of informal businesses. Places with kiosks and stalls tend to become inappropriate disposal points, where the accumulation and abandonment of waste in the Solarium Area is more frequent (Araújo et al., 2012). In the region with the least development (CP), the presence of these facilities is smaller and generates less waste on the sand strip.

This information also influences disposal in different areas of the beaches, as the Solarium Area recorded a total value of 1.66 cigarette butts/m² in BV, 1.26 in PG and with a density of 0.28 cigarette butts/m² in CP. In the High Tide Line Area BV had a higher value of 3.24 cigarette butts/m², while PG recorded 2.57 cigarette butts/m² and CP a total of 0.42 cigarette butts/m², with only the last beach recording a significant difference ($p < 0.05$) in both the Solarium and the High Tide Line Area. The High Tide Line Area does not usually have many smokers, but cigarette butts can be discarded by passers-by or carried by the wind (Araújo and Costa, 2019), and the presence of collectors in this area can be limited, making rubbish bin installations a challenge.

Boa Viagem and Porto de Galinhas, although they have different levels of development and occupation, share the same rate of cigarette butt pollution. However, Cupe beach, even with its low level of use and development, had a pollution level classified as moderate (Table 4). Both municipalities are responsible for tourism activities in the local and state economy, with Recife leading the ranking in terms of Gross Domestic Product (GDP), while Ipojuca is in third place in terms of GDP for the state of Pernambuco (IBGE - Instituto Brasileiro De Geografia E

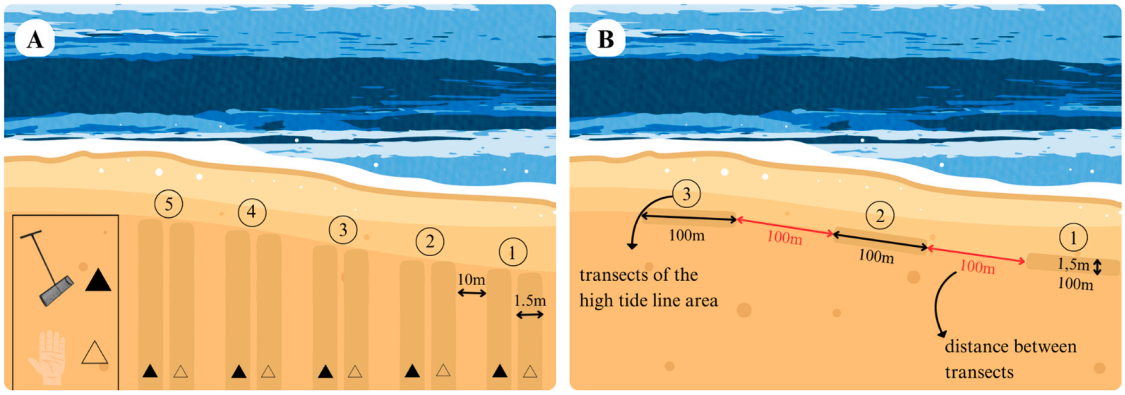


Fig. 2. Collection methods applied in the solarium region, indicating the dimensions of the transects (1.5 m) and the spacing of 10 m between them, as well as the collection method used in the leave line, with its respective transect dimensions and distance between sampling points. Source: Nirhvana Felipe, 2025.



Fig. 3. Four levels of cigarette butt degradation established by Araújo et al. (2022).

Table 3
Reference values according to the studies used to determine the cigarette butt pollution index in the study areas.

REFERENCE	REFERENCE VALUES
Torkashvand et al. (2021)	Very low pollution = 1 or less Low pollution = 1.1–2.5 Moderate pollution = 2.6–5 Significant pollution = 5.1–7.5 High pollution = 7.6–10 Severe pollution = above 10
Alkalay, Pasternak and Zask (2007).	Very clean = 0–0.1 items/m^2 Clean = 0.1–0.25 items/m^2 Moderate = 0.25–0.5 items/m^2 Dirty = 0.5–1 items/m^2 Extremely dirty = more than 1 items/m^2

Estatística, 2025). This information may be directly linked to the type of service offered in these regions, since Recife and Ipojuca have different occupancy figures in terms of population/km^2.

This study shows a contamination index that tracks contamination levels at different beaches around the world. Nguyen et al. (2025) observed a CBPI on beaches in Vietnam with contamination levels ranging from "very low" to "low," while Yona et al. (2024) indicated 'low' to "severe" levels on beaches in East Java province. Approximately 45 % of the beaches evaluated were classified as having low pollution, while 18 % had significant pollution and another 18 % were considered highly or severely polluted (Mghili et al., 2023).

As observed by (Araújo et al., 2012), waste discarded on the beach spreads and can be buried and/or carried away by the tide, causing interaction (Fig. 5A) or even ingestion by marine species, affecting biodiversity and entering the food chain. These figures may also be influenced by the reduction in enforcement or differences in the behaviour of tourists/users in relation to rubbish disposal on the different beaches. The main problem is proper disposal, as cigarette butts are not easy to store in pockets or purses until a rubbish bin is available. It is necessary to implement public management measures that reconcile environmental conservation with economic development, especially in dynamic environments such as the coastal zone.

Another factor that influenced the density of cigarette butts on beaches was the collection period, in which the month with the highest rainfall showed a reduced amount of material, with different seasons potentially influencing the number of items (Dantas, Mendes and Gentil, 2025). The rainfall values in Fig. 6 express this rainfall information in relation to the collection days, where the days during the rainy season (March) together showed a rainfall of 4.8 mm, the rainy season (March) with 34.1 mm, the dry season (September) with a total of 1.8 mm and the dry season (December) showing the lowest rainfall with 0.2 mm.

Between the rainy and dry periods, it was possible to identify a significant difference ($p < 0.05$) in the number of cigarette butts/m^2 in the three study areas, also considering this difference in the Solarium and High Tide Line Area and in the collection methods (beach cleaning equipment and manual). Through Pearson's correlation (Fig. 6) it was possible to corroborate this information through the correlation coefficient values (r). In the Solarium Area, the beach with high development ($r = -0.79$) and medium development ($r = -0.76$) showed a moderate negative correlation, while the beach with low development ($r = -0.28$) showed a very weak negative correlation. In the High Tide Line Area, it was observed that the beach with high development ($r = -0.96$) had a very strong negative correlation, while the beach with medium development ($r = -0.87$) and low development ($r = -0.86$) had a strong negative correlation. This highlights that the density of cigarette butts is inversely proportional to rainfall, especially in the High Tide Line Area on beaches with high and medium development.

Based on this hypothesis, it is assumed that the distribution of cigarette butts is associated with an increase in their occurrence during the dry season, compared to the rainy season in beach areas, due to the higher frequency of tourists and movement of users (Silva, Araújo, and Silva-Cavalcanti, 2023b). In contrast, during the rainy season, in addition to the drop in the number of people in these regions, there is also the factor of reduced efficiency in capturing this material, as it camouflages itself (Fig. 5B) in the sediment when it is soaked with rainwater. Rainfall also influenced the collection method, as during the rainy periods (March and June), the density of cigarette butts collected by beach cleaning equipment and by manual capture was quite similar, unlike during the dry months (September and December). There was no significant difference ($p > 0.05$) between the collection methods, and beach cleaning equipment could be one of the possible alternatives for capturing this material.

Hand collection is more efficient; however, it takes much more effort when compared to equipment, and beach cleaning equipment can be a

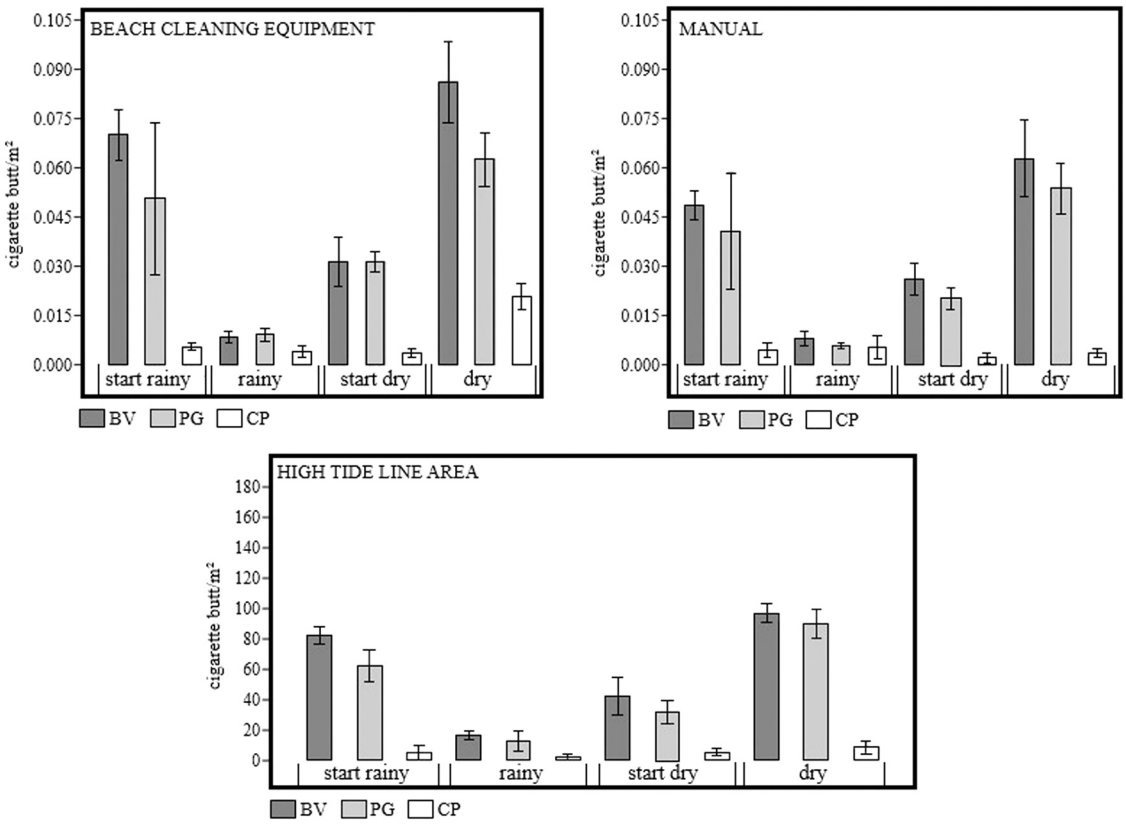


Fig. 4. Cigarette butt/m^2 in the Solarium and High Tide Line Area, comparing collection methods in the Solarium Area (beach cleaning equipment and manual) at different times of the year on beaches with a high level of development (BV), medium level of development (PG) and low level of development (CP).

Table 4
Index of pollution of beaches by cigarette butts, using as a reference the calculations of the studies indicated in the methodology, based on the density of cigarette butts found on the beach with a high level of development (BV), medium level of development (PG) and low level of development (CP).

REFERENCE	BV	PG	CP
Torkashvand et. al. (2021)	Severe pollution	Severe pollution	Moderate pollution
Alkalay, Pasternak and Zask (2007)	Extremely dirty	Extremely dirty	Moderate

good alternative for reducing cigarette butts (or similar sized materials) on beaches during dry periods. However, it is important to note that the adoption of a set of measures is essential, because in periods of high rainfall, the compaction of the sediment may have compromised the efficiency of the beach cleaning equipment. During collection, at times the equipment became clogged, making the process difficult and possibly leading to underreporting of the waste collected, and during periods of high rainfall manual collection also required a great deal of effort. As a result of this information, it is essential to combat new cigarette butt disposal, since there is currently no alternative, efficient equipment for removing cigarette butts in rainy periods, and manual collection is not advisable due to the effort involved, which could lead to health problems for cleaning workers.

This equipment has been used as an alternative to enhance the cleaning service in these areas, mainly to remove cigarette butts or other materials with similar sizes that are ignored during traditional cleaning (Widmer and Reis, 2010). A public agency in the state of Alagoas in Brazil has already incorporated this method for removing cigarette butts, beer caps, can seals and plastic fragments. The effectiveness of the equipment in reducing the reduced sizes on the beaches of Maceió-AL has been validated through tests (Prefeitura De Maceió, 2025), demonstrating its potential for application in other coastal regions facing the same problem. These measures are important because the conservation and sustainability of the oceans, seas and marine resources is one of the pillars of the United Nations' SDG-14.

Another factor observed was the density of cigarette butts according to the transects, as locations with more infrastructure and services (bars, street vendors, chairs, and hotels) had a higher number of cigarette butts, which was also observed in the study by Díaz-Mendoza et al., (2023). It was observed that there was no significant difference between the transects in any of the study areas ($p > 0.05$), and it can be emphasized that regardless of the level of development of the beach, if the site has services and structures that influence agglomerations associated with the absence of cleaning and lack of collectors, it will trigger the increase and accumulation of cigarette butts, and areas with availability of these services can be determined as critical points for the ecosystem. Once these critical points have been identified, it is necessary to reinforce them with a set of measures to raise awareness among users, traders and stallholders, as well as the implementation of efficient measures by the executive body.

Although Decree Nº. 8.262/2014 has been in force for 11 years, banning the smoking of cigarettes and tobacco products in public or private places of collective use in Brazil, it is not applied to open areas such as beaches (Brasil, 2014). International models can be used as a reference, such as Mexico, which has banned smoking in public places (El País, 2025) and New Zealand, which has passed a law setting a minimum age for smoking, with the aim of banning smoking for generations to come, as well as reducing its irregular disposal (The Guardian, 2025). These measures could be extended to countries such as Brazil, Iran and Slovenia, which have a severe level of cigarette butt contamination in public places (Silva, Araújo and Silva-Cavalcanti, 2023b). However, the establishment and enforcement of laws and decrees does not exempt new waste, and effective monitoring of these

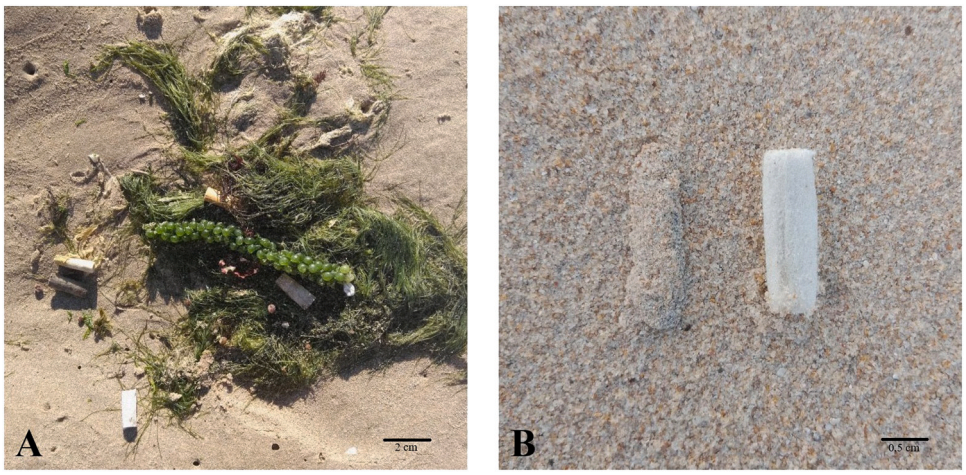


Fig. 5. Cigarette butts found on the beach with the highest level of development (BV). Image A, captured in the High Tide Line Area, shows a seaweed entangled with cigarette butts. Image B, taken on the strip of sand, shows the camouflaging potential of this waste, with one cigarette butt completely covered by the grains of sediment and the second completely exposed. Source: Nirhvana Felipe, 2022.

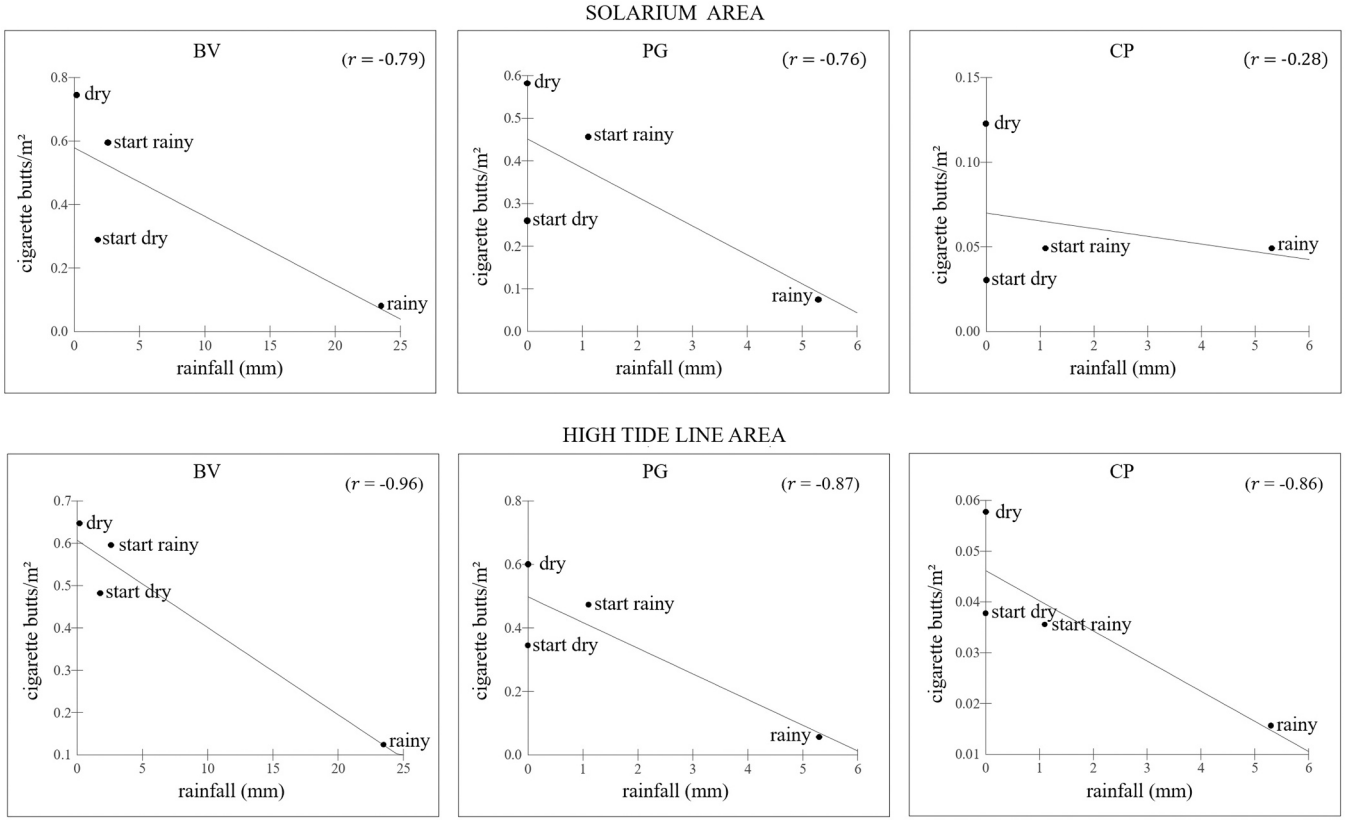


Fig. 6. Pearson's correlation (with correlation coefficient values = *r*) between rainfall and cigarette butt density/m² on beaches with high (BV), medium (PG), and low (CP) levels of development, considering the Solário regions (using both collection methods) and the High Tide Line Area, considering different periods of the year.

sites is also necessary.

The transect values also reflect the levels of degradation found at the study sites. The Principal Component Analysis (PCA) showed that the recently discarded cigarette butts (Type I) are mainly found in the transects where there are the greatest movement and permanence of users both on the sand strip and in the area, which is the equivalent of the first (T1) and second (T2) transects of the High Tide Line Area shown in Fig. 7.

The cigarette butts found in the Solarium Area showed a higher density of material in an advanced stage of degradation in the transect furthest from the concentration of users, the fifth transect (T5), which may be related to their accumulation power and the low efficiency of the cleaning system in removing this item or materials of a similar size. According to the study conducted by Díaz-Mendoza et al. (2023), different microorganisms of the genus *Virgibacillus* were identified, which is relevant for its potential in bioremediation, in addition, total coliforms and *Escherichia coli* were detected in sand, with a higher presence in the active and resting zones.

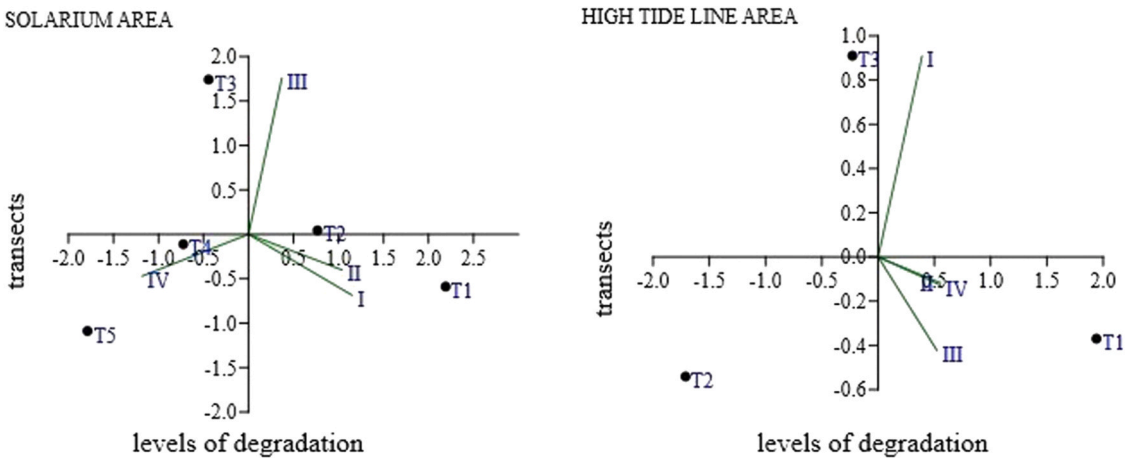


Fig. 7. Principal Component Analysis (PCA) between levels of cigarette butt degradation (I, II, III and IV), with the five transects of the Solarium (T1, T2, T3, T4 and T5) and the three transects of the High Tide Line Area (T1, T2 and T3) of the areas collected.

It was also possible to identify the density of material degradation according to the beaches and the Solarium and High Tide Line Area (Fig. 8). The study areas in this study showed the 4 levels of degradation, with a significant difference ($p < 0.05$) for cigarette butts in an advanced stage of degradation (type IV), which means that these cigarette butts may have remained on the beach for approximately two years (Bonanomi et al., 2015). In the High Tide Line Area, the accumulation of type IV cigarette butts was observed, with a higher density on the more developed beach. As for the medium and low development beaches, the value of type IV cigarette butts was lower, with higher values than the other types I, II and III for the same area (Fig. 8). This relationship may be associated with the physical characteristics of the beaches, where medium and low development beaches have a smaller strip of sand, leading users and street vendors to occupy the High Tide Line Area more intensely.

This long period of exposure encourages the material to wear down and can lead to the release of fibres through friction with sand (Araújo et al., 2022). This material, which is in an advanced stage of degradation, is also becoming increasingly smaller and therefore more noticeable, increasing the possibility that it will be mixed in with the sand and making it more difficult to remove, which reinforces the need for effective methods for detecting and collecting it. In addition, by

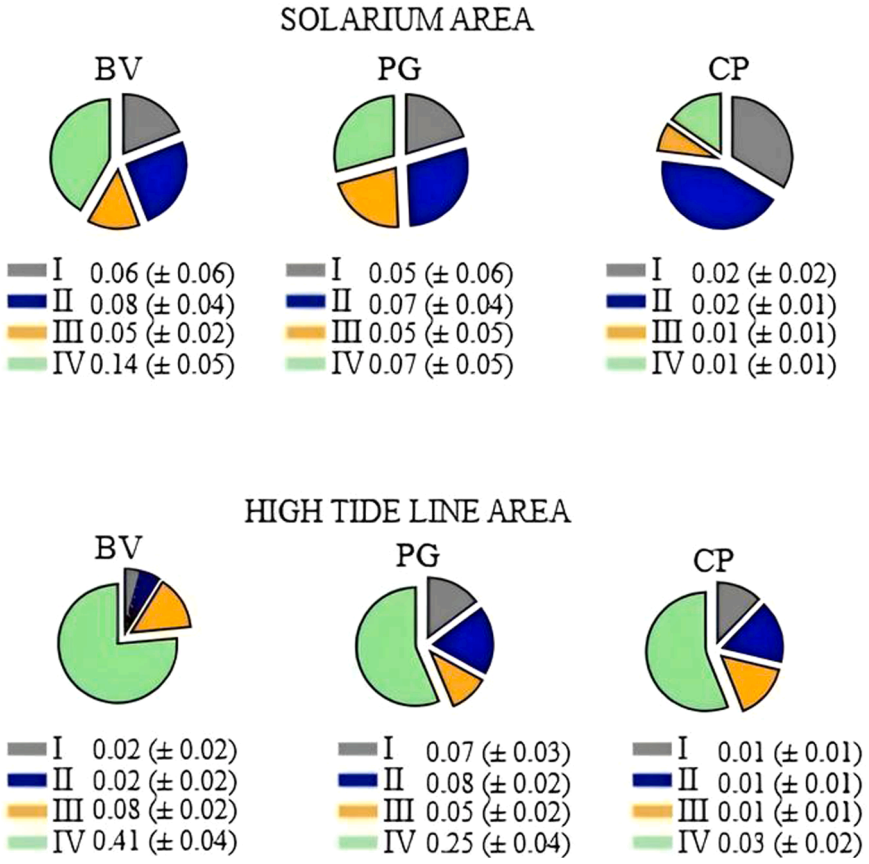


Fig. 8. Density of degradation levels of cigarette butts found on the beach with high level of development (BV), medium level of development (PG), and low level of development (CP), considering the area of the Solarium and the high tide line.

reaching this level of deterioration, it is likely that the compounds present in cigarette butts have already been released, contaminating the environment over the years (Bonanomi et al., 2015). In this study, we found that the High Tide Line Area of the beach with the highest level of development (BV) had a higher level of contamination when compared to the beaches with medium (PG) and low development (CP). This may be influenced by several factors, including the lack of cleaning in this area of the beach and the burial of cigarette butts by the waves and trampling by people.

The visual condition as well as the loss of mass of cigarette butts can help to highlight the seriousness of the problem related to this toxic material, as the filaments of cigarette butts are released. For comparison, the pattern of reduction observed in this study was even more pronounced than that reported by Silva et al., 2023b, who recorded a reduction of around 40 %, while the cigarette butts of types I and IV collected on the three beaches showed a reduction of approximately 52 % of the total mass (Table 4). There was a significant difference ($p < 0.05$) in the weight of the material between types I and IV. Type I had an average of 0.17 g (±0.09); Type II: 0.16 g (±0.07); Type III: 0.13 g (±0.04) and Type IV: 0.08 g (±0.02), but there were no significant differences ($p > 0.05$) in morphometry and weight between the beaches. However, from these values, it was possible to identify that the dimensions of the material are restricted to mesolayers (5 mm to 2.5 cm) and we can assume that the beach with the highest development index (BV), because it has a lower weight of material (Table 5), represents the beach with the greatest potential for micro-fragments from cigarette butt wear.

This information is fundamental to understanding contamination in coastal environments, since cigarette butts contain highly toxic chemicals such as arsenic, benzene, cyanide, polycyclic aromatic hydrocarbons (PAHs) and heavy metals. According to Silva et al., 2023a, ecotoxicological tests carried out with cigarette butts showed sublethal effects on organisms and also identified the word “mortality” with the highest significance value among the publications analysed. In addition, Torkashvand et al. (2021) stresses that a single cigarette butt can pollute up to 1000 litres of water, and the environmental impact can vary according to the type of cigarette discarded, since different cigarette compositions (white and red), filters (acetate, paper and biodegradable) and additives (mentholated and with essence) can increase the potential for pollutants in the environment. Through studies with mentholated cigarettes, it was possible to identify impacts on germination and biomass in *Lolium perenne* L. and *Trifolium repens* L (Green et al., 2019), in addition to observing the high teratogenic potential of mentholated cigarette butts causing malformation in Xenopus laevis (Parker and Rayburn, 2017).

In addition, the diversity of cigarette butts (types of brands) is also an important factor in identifying their environmental impact, since, depending on the cigarette’s manufacture, it can generate different levels of toxicity. A study by Osuala et al. (2017) identified impacts on the liver and gills of tilapia, with results varying according to the diversity of brands tested. This type of analysis contributes to a better understanding of the environmental damage associated with the improper disposal of cigarette butts and helps to develop more effective strategies for their control. The identification of brands is also essential to determine the prevalent polluters, enabling more targeted actions to be taken to hold them accountable and mitigate the pollution caused by this waste.

Table 5
Morphometric characteristics of cigarette butts collected on a beach with a high level of development (BV), medium development (PG) and low development (CP), considering the Solarium and High Tide Line Area.

LOCATION	WEIGHT (g)	LENGTH (mm)	WIDTH (mm)
Solarium Area			
BV	0.15 (± 0.07)	23.09 (± 8.77)	7.14 (± 1.52)
PG	0,17 (± 0.06)	23.40 (± 3.51)	7.60 (± 1.10)
CP	0.18 (± 0.09)	23.89 (± 3.69)	7.61 (± 1.34)
High Tide Line Area			
BV	0.13 (± 0.07)	22.43 (± 3.43)	6.79 (± 1.49)
PG	0.18 (± 0.06)	22.80 (± 8.64)	7.11 (± 1.42)
CP	0.14 (± 0.04)	22.71 (± 2.97)	6.93 (± 1.23)

In this study, 1012 cigarette butts were analyzed, of which 38 % contained brand identification, and a total of 16 brands were found (Fig. 9). Of the total observed, three marks with the highest values (A, B, C and D) accounted for 33,7 % of the pollution in the study areas of this work, where these same marks were also found on the beaches of Recife in 2022 (Silva et al., 2023b). The audit identified that, in the locations of BV and PG, the predominant brands were A, B and D, whose sale prices are established by Agência Nacional De Vigilância Sanitária (Anvisa), 2025b as follows: brand A at R$ 9.25 (US$ 1.74), brand B at R$ 12.00 (US$ 2.25) and brand D at R$ 11.75 (US$ 2.20). On CP beach, brands A, B and C stood out, with brands A and B following the same price list, while brand C has a lower price of R$ 6.50 (US$ 1.22). In several regions, the low purchasing power of the population favours the choice of more affordable cigarette brands (Araújo et al., 2012). These prices may encourage consumption and, consequently, the improper disposal of cigarette butts on beaches (Silva et al., 2023b). This information reinforces the urgent need for public policies that integrate price regulation, differentiated taxation and effective waste management strategies.

By consulting the Register of Smoking Products Derived from Tobacco, made available by the National Health Surveillance Agency (Agência Nacional De Vigilância Sanitária Anvisa, 2025a), it was possible to identify 3 that are sold illegally in Brazil. Among the unregistered brands, the “L” brand, which is produced in Argentina, and the “M” brand, manufactured in Spain and Belgium, were found on the beach with the highest development (BV), while the “P” brand is produced in the United States and was found on the beach with medium development (PG). This may indicate that these materials did not pass an inspection during the user’s entry into the country or that these cigarettes may have entered through illegal sales.

The weakness of the controls and the slowness of the judicial process lead to illegal trade in some countries, such as Brazil, where around 9 billion illegal cigarettes are produced every year (Ramos, 2009). Despite ANVISA’s efforts to analyze the components of cigarettes available on the market, many still enter the country illegally, posing an even greater risk. As these products are not controlled during production, they can contain a much higher number of compounds, as well as a greater potential load and a variety of potentially higher compounds than legalized cigarettes.

The tobacco products trade cannot absolve itself of responsibility for the impacts caused by the irregular disposal of cigarette butts, and companies can include the UN’s SDG 12, aimed at the sustainable production and consumption of these materials. According to a study by Ribeiro et al. (2025), two cigarette companies were responsible for 57.6 % of the more than 4000 cigarette butts collected, and should be held responsible for waste management costs. Furthermore, in Spain, a law has been implemented that transfers responsibility for the cost of cleaning up cigarette butts discarded in public spaces to manufacturers (BBC, 2025), which could be an effective strategy for mitigating this environmental problem in coastal regions.

Some companies are willing to alleviate the problems of cigarette production and are looking for alternatives to sell new materials. Like a tobacco products company that hired a former UN human rights defender to act as Director of Social Impact and Sustainability, seeking to promote joint efforts between the private sector and NGOs to transform the industry towards sustainability, with a reformulated business model and a focus on other products (Motles, 2024). However, few efforts are being made to deal with materials that are already present in the environment, without effective treatment or reduction of their impacts.

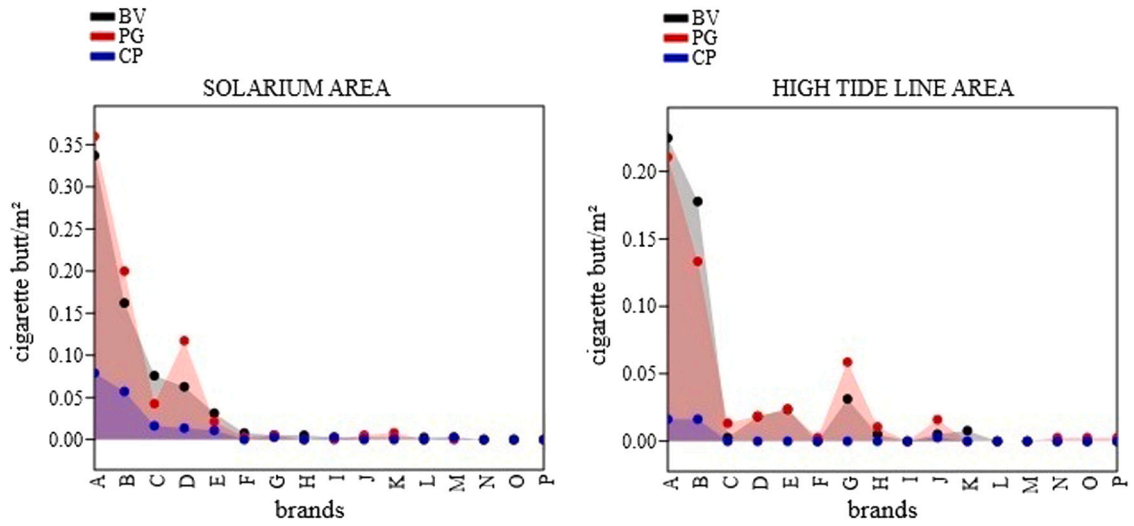


Fig. 9. Distribution of the main pollutants prevalent on beaches with high levels of development (BV), medium levels of development (PG), and low levels of development (CP) based on the density of the marks identified, recorded in hidden form using the letters "A" to "P.".

4. Conclusion

Cigarette butts should be adopted as flag items in assessing beach contamination through density indices. The growing pollution of beaches by cigarette butts calls for innovative solutions for their collection and disposal. The study revealed that cigarette butt contamination is more intense on beaches with greater urban development, such as Boa Viagem and Porto de Galinhas, and that its distribution varies according to seasonality and rainfall, being higher in the dry season due to increased tourism.

The degradation of cigarette butts contributes significantly to environmental pollution, especially in areas where the density of this waste is higher, such as the High Tide Line Area. In addition, beach cleaning equipment showed greater efficiency in capturing cigarette butts compared to manual collection in the Solarium Area, suggesting its potential as a tool for the sustainable removal of this waste.

To tackle this problem, it is essential to adopt effective cleaning strategies, such as beach cleaning equipment that is both efficient and sustainable and applicable to the sandy strip of beaches with different levels of development. However, in addition to investing in new technologies, it is necessary to promote environmental awareness among bathers and tourists, as well as public policies such as the polluter pays principle, which makes the producer responsible for the final disposal of the product.

Given the severity of the pollution caused by cigarette butts, it is essential to advance the debate beyond mitigation measures on beaches and in urban areas, incorporating alternatives within the industrial sector itself. Among these, the development of fully biodegradable cigarettes and the progressive reduction of production stand out, as these possibilities could be long-term structural solutions capable of addressing the root of the problem and promoting effective change in reducing waste with high contamination potential.

Contributorship statement

Nirhvana Felipe da Silva: Processed the experimental data, performed the analysis, drafted the manuscript; and Jacqueline Santos Silva-Cavalcanti: Planning, conceived of the presented idea, verified the analytical methods.

CRediT authorship contribution statement

Jacqueline Santos Silva-Cavalcanti: Writing – review & editing, Validation, Supervision, Project administration, Data curation. **Silva Nirhvana Felipe:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.rsma.2025.104676.

Data availability

Data will be made available on request.

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